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PI Name:	O'Donnell, Janis M Ph.D.		
Project Title:	Catecholamine regulation and response to environmental change in Drosophila melanogaster		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Human Research Program Risks:	None		
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Space Biology Special Category:	None		
PI Email:	jodonnel@bama.ua.edu	**Fax:**	FY 205-348-1786
PI Organization Type:	UNIVERSITY	**Phone:**	205-348-9810
Organization Name:	University of Alabama		
PI Address 1:	Box 870344		
PI Address 2:	Department of Biological Sciences		
PI Web Page:			
City:	Tuscaloosa	**State:**	AL
Zip Code:	35487	**Congressional District:**	7
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	This project utilizes the genetic model organism, Drosophila melanogaster, the fruit fly, to explore the response of multicellular organisms to extreme environmental change and to identify factors that determine or modify the ability of this model system to tolerate environmental change. This work has focused primarily on physiological pathways that are known to be responsive to stressful conditions in mammals, pathways which utilize the neurotransmitter and neurohormonal family of chemicals, the catecholamines. These are highly conserved pathways. Very similar genes and protein products of these genes are found in all organisms with nervous systems. The project has been based on three hypotheses. The first was that catecholamine production is regulated in response to environmental stress in Drosophila as it is in mammals – an essential criterion if Drosophila is to be a useful model. The second was that regulation of the catecholamine pathway is critical to an organism's ability to tolerate stressful environments. This hypothesis is one that		

Task Description:	the Drosophila system is uniquely suited to address since a large collection of strains with mutations in the critical genes makes possible the testing of physiological consequences for stress tolerance in strains with perturbed catecholamine regulation. The third hypothesis was that gravitational change, even though it is not an environmental condition normally encountered by Drosophila, would trigger responses that are similar or identical to those caused by a variety of stressors that are encountered in its normal environment. The three key aims in this project were: 1) To determine whether mutation-induced changes in catecholamine regulation affect ability to tolerate environmental change, aging, behavior and fertility. (2) To determine the molecular genetic responses of catecholamine-regulating genes to environmental changes. (3) To identify cell types and developmental stages most responsive to and affected by environmental change. Space travel subjects humans and other organisms to extreme environmental change. It is critical to understand cellular mechanisms and genetic variation that can dictate the tolerance of organisms to such change. We have identified genes in Drosophila whose normal function and regulation is crucial for the ability to cope with stress, and these genes are virtually identical in sequence and function to their human counterparts. Moreover, we have found that dietary supplementation with the products of the pathways in question can modify the ability to withstand environmental change. Use of Drosophila as a model system for this study allows the application of a powerful array of genetic and molecular technologies, which in this instance, have provided insights into genetic variation that directly modifies the ability of this organism to cope with environmental extremes. The high degree of genetic conservation among multicellular organisms means that it is very likely that our results can be extrapolated to higher organisms. The wealth of genetic materials, the ease of rearing vast numbers of organisms in very small spaces, and the low cost of producing and maintaining mutant strains also contribute strong arguments for pursuing these studies in this model. We have found evidence supporting the hypothesis that dopamine regulation affects tolerance to environmental stress, a finding that has implications for the well-being of animals and humans in space travel.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A key feature of the ability of animals to convert perception of environmental change into appropriate metabolic responses is the system that regulates the synthesis of catecholamine neurotransmitters and neurohormones. These responses allow organisms to adjust to new environments, to make protective physiological changes when harmful environments are encountered, and to initiate appropriate behavioral changes. Regulation of the production of catecholamines is particularly evident under conditions of extreme environmental variation. While these metabolic responses allow organisms to tolerate quite disparate environmental conditions in many instances, they can also have negative consequences on health and reproduction. Such consequences are due to the diverse signaling functions that catecholamines have in all animal species. In vertebrates, they have been implicated as signaling molecules in cardiac development and in both vertebrates and insects, roles in ovarian development have been observed. Perturbations in catecholamine levels cause abnormalities in sensory development and function, in neuromuscular and motor neuron activity and in certain reproductive behaviors in Drosophila. It is therefore crucial to identify the molecular components of the catecholamine pathway that participate in regulating responses to environmental change and to ascertain the physiological consequences of the catecholamine response in more detail. Our work this year has lead to several striking and unexpected results that have important implications with respect to the long term effects of stress because some of the changes that we observe in our mutants, changes that enhanced by stress, are molecular changes that may be implicated in addictive behaviors and others in the progress of Parkinson's Disease. In conjunction with these studies, we have developed a neurodegenration model which will allow us to survey the neural effects of potentially deleterious environmental agents. We anticipate that these studies will lead to strong mechanistic descriptions of how this organism responds and adjusts to environmental conditions. We have obtained a better understanding of the role that the catecholamine pathway plays in physiological alterations associated with flight conditions. Because this pathway and the genes involved are highly conserved, it is expected that this study will provide more generally applicable insights into all animal species' responses to environmental change.
Task Progress:	We have completed work on all specific aims for this project. The key results stemming from this research are as follows. 1.Mutations in dopamine regulating genes, as hypothesized, modify stress tolerance and longevity. 2.These effects are observed in heterozygotes carrying one wild type copy of the gene, which otherwise exhibit little evidence of effects from the mutant alleles. That is, in very case, modification of stress tolerance was a dominant phenotype. This is of significance as it implies that cryptic genetic variation can alter responses and tolerance under conditions in which environmental conditions change. 3.Environmental changes alter expression of dopamine pathway components and the size of dopamine pools in the central nervous system. 4.Changes in stress tolerance observed in heterozygous mutant backgrounds were virtually identical for each stress condition assayed: hypergravity, starvation, desiccation, oxidative, and heat. 5.In all cases, males were consistently more sensitive to altered environmental conditions than females. 6.Employing a Green Fluorescent Protein reporter expressed in dopaminergic neurons, it was found that oxidative stress conditions resulted in degeneration of these neurons. However, it appears that acute starvation and desiccation stress do not affect dopaminergic neuron survival, nor architecture at a gross morphological level. 7.Mutations in the gene Catsup are neuroprotective, when the adult fly is exposed to oxidative stress via a known neurotoxin.
Bibliography Type:	Description: (Last Updated: 08/06/2008)
Abstracts for Journals and Proceedings	O'Donnell, J.M.; Chaudhuri, A.; Izevbaye, I. "Tetrahydrobiopterin and dopamien signaling pathways modulate oxidative stress response in Drosophila melanogaster." Internationation Symposium on the Chemistry and Biology of Pteridines and Folates. Pteridines 16: 66 [009-04] , Jun-2005

Abstracts for Journals and Proceedings	Chaudhuri, A.; Brown, J.; O'Donnell, J. "Dopamine pathway modulation of oxidative stress response in Drosophila melanogaster." 46th National Drosophila Conference, San Diego, March 2005 http://www.faseb.org/genetics/dros05 , Feb-2005